Evaluating the Impact of Electrophysiologic Cardiac Sex Differences in Vulnerability to Antidepressant-Induced Arrhythmias.

Candidate 1091797

Abstract

While antidepressant prescriptions have tripled in 20 years, their cardiac ion channel block, which can trigger arrhythmias such as Torsades de Pointes, remains under-evaluated, particularly for women whose reduced baseline I_{Kr} elevates risk not accounted for in current sex-neutral models and prescribing guidelines. In this study, we leverage ToR-ORd to quantify how electrophysiological sex differences affect vulnerability to arrhythmias induced by sertraline (SSRI), amitriptyline and desipramine (TCAs). We find that female models develop early after-depolarisations at therapeutic doses whereas male models do not; sertraline shows the greatest risk followed by amitriptyline, while desipramine remains EAD-free due to compensatory I_{CaL} block. Mapping a simulation-derived risk threshold onto PTB-XL ECGs, 20% of women fall into the elevated-risk category compared to only 14% of men. These results provide sex-differentiated evidence of quantifiably higher antidepressant-induced arrhythmic risk for women, supporting the need for more personalised and sex-differentiated prescription guidelines.

1. Introduction

From 2011 to 2022, England experienced a substantial increase from 5.8% to 13.0% in the prevalence of depression, propelled by COVID-19 and stabilising above pre-pandemic levels [1]. This figure also reveals that women are 39% more affected than men, a significant sex gap.

Along with that rise, prescriptions for antidepressants have tripled in English primary care over the 1998-2018 period, a market dominated by selective-serotonin reuptake inhibitor (SSRIs) and tricyclic antidepressants (TCAs) [2]. These drugs block cardiac ionic currents and expose patients to QTc prolongations that can trigger arrhythmias such as Torsades de Pointes (TdP) [3].

1.1. Research question

With anatomical differences lowering I_{Kr} by 18%, women also display naturally longer QTc intervals which further increase the risks of arrhythmias [4]. There, a relevant question arises: how do electrophysiological cardiac sex differences affect vulnerability to arrhythmias induced by antidepressants? Additional risk coming from sex differences could be quantified by biomarkers prior to treatment and considered carefully when prescribing antidepressants. To address this question, our study examines three compounds: the sertraline SSRI, the most prescribed antidepressant, alongside amitriptyline and desipramine, two commonly used TCAs whose proarrhythmicity has been examined without sex differences considerations [2, 5].

1.2. Computational methods

On top of the danger of administering patients with arrhythmogenic doses of antidepressants, clinical studies cannot isolate cardiac channel mechanisms as well as ventricular cell models. In addition, animal studies have been shown to underperform computational methods when considering an *in silico* population of models approach [6]; while clinical trials are limited by ethical and practical constraints, computational methods generate scientifically grounded cardiomyocytes with controlled variability [7]. Although this approach omits tissue-level effects that also affect arrhythmia, it appears ideal to study ion channel mechanisms and define risk levels for antidepressants.

1.3. Aims and background

This study seeks to leverage ventricular cell models to quantify the impact of sex differences on arrhythmic thresholds for sertraline, amitriptyline, and desipramine, enabling the characterisation of sex differences and risky ion channel blocking profiles. This study then aims to evaluate its findings against the PTB-XL ECG dataset to incorporate our findings to a relevant clinical setting.

Contextualising our study, existing literature establishes higher drug-induced TdP risk for women, despite no universal scientific explanation [8]. In parallel, prior work quantifies the ionic channel blocks for sertraline, amitriptyline, and desipramine [6, 9, 10], revealing that antidepressants have arrhythmogenic effects categorised as high as Class II (probable TdP risk) [5]. Our research thus addresses a gap in the literature by considering a sex-specific cardiac models for antidepressants and applying our findings to real-world data [4, 7].

2. Methods

We use ToR-ORd on MATLAB_R2025b to represent cardiomyocytes by capturing depolarisation, repolarisation, and calcium dynamics properties [7]. We combine it with a population of models approach to detect minorities absent in average patient simulations. Following prior work, we generate 30 samples per sex with Latin Hypercube Sampling over seven ion channel conductances [6].

To apply our results on existing ECGs, we then use PTB-XL, a public dataset of 21,837 annotated 12-lead ECGs [11]. We notably filter for normal ECGs (NORM confidence $\geq$ 80) and perform an analysis on the remaining 8,798 records using Python 3.11.

2.1. Foundation of models and parameters

ToR-ORd follows the Hodgkin-Huxley framework, where ionic currents are modelled as conductance-gating products and drug effects are captured by the Hill equation. Sex differences are introduced by scaling six ionic conductances in line with existing literature; I_{Kr} is notably reduced by 18% in comparison with male models [4]. Finally, to bridge single-cell results to clinical ECGs, we map simulation from APD_{90} to QTc without assuming any direct scaling since APD_{90} measures single-cell repolarisation and QTc whole-heart repolarisation.

2.2. Experiment protocols

We start by performing simulations for sertraline, amitriptyline and desipramine on ToR-ORd. After establishing ranges of doses using public labels backed by clinical references [9, 12, 13], we check if average male and female patients show signs of early afterdepolarisation (EAD) at maximum doses for varied basic cycle length (BCL). Plasma concentrations are converted from mg/day to μM as $\frac{\text{Dose (mg/day)} \times 1000}{24 \times CL/F \times MW}$, where CL/F is apparent oral clearance and MW is molecular weight.

Following this, a population of models simulates exposure to antidepressants across sexes. For each drug, we fix BCL = 1000 ms before recording the lowest concentration and APD_{90} triggering EADs (if any). We then compare arrhythmia exposure for each sex and compare results for BCL = 600, 1000, and 1200 ms to observe the impact of I_{Kr} block at different heart rates.

Lastly, we correlate female normal and drug-induced APD_{90} on PTB-XL to see if baseline repolarisation duration predicts arrhythmic vulnerability under drug exposure. From this analysis, we define a simulation-derived risk boundary mapped to a QTc threshold against the female PTB-XL distribution.

To ensure open-source and reproducibility, our simulations use publicly available data and frameworks. IC_{50} values and Hill coefficients are thus taken from published literature, our ECG data is publicly available, and a random seed is fixed to enable reproducible LHS, [9–11].

2.3. Rejected approaches

Several modelling approaches were considered and ultimately rejected. For example, tissue-level AP propagation simulations using MonoAlg3D was too complex to yield interpretable results. Since it required a 3D cardiac mesh, fibre orientation field, heterogeneous transmural cell parametrisation, and conduction parameters, these additional degrees of freedom could easily confuse genuine sex difference from artefacts of the geometrical assumptions.

Training a machine learning risk classifier is another candidate limited by the lack of labelled ECGs including drug-exposures. Since no such public dataset exists, predictions derived from simulation thresholds would be impossible to validate clinically, preventing the definition of a loss metric. MIMIC-IV stands out as a strong candidate for this task, but its credentialed access and complexity require substantial feature engineering and domain knowledge, limiting such projects to established medical labs.

3. Results and Discussion

3.1. Experimental results

After comparing all three drugs on average male and female patients at varying heart rates for maximum therapeutic doses without any EAD, we simulate a population of 30 models with Latin Hypercube Sampling for each drug.

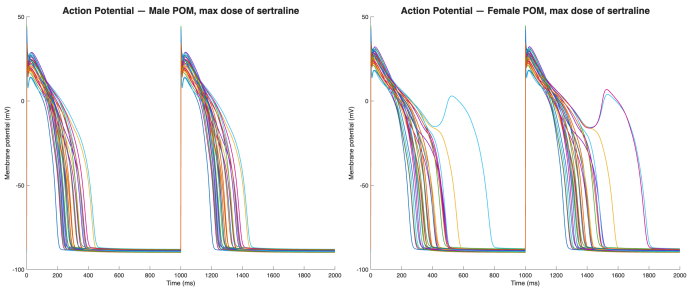

Figure 1: AP of men (left) and women (right) using a population of models for a maximum therapeutic dose of sertraline and a BCL of 1000 ms. Sex-driven additional arrhythmia risk is evident from the EADs observed in a subset of the female population, which are absent in men.

For a maximum therapeutic dose at 60 BPM, 2 female models trigger EAD for sertraline, 1 female model triggers

EAD for amitriptyline, and all other AP simulations appear normal. Changing the number of population models, our results scale proportionally. The female EAD threshold for sertraline is 0.110 μM, revealing cardiotoxicity for vulnerable women in most of the upper therapeutic range. Amitriptyline triggers EADs in women from 0.445 μM, almost at maximum dose.

These results suggest that arrhythmia is driven by I_{Kr} block, as sertraline's lower I_{Kr} IC_{50} achieve the strongest fractional hERG block at clinical concentrations. Nonetheless, desipramine triggers no EADs despite stronger I_{Kr} block than amitriptyline, showing that other channels also modulate risk; I_{CaL} block reduces the inward current that sustains plateauing and triggers EAD. This result thus supports the claim that QTc prolongation is not sufficient to predict arrhythmic risk: desipramine produces 73% greater APD_{90} prolongation than amitriptyline yet remains EAD-free [14].

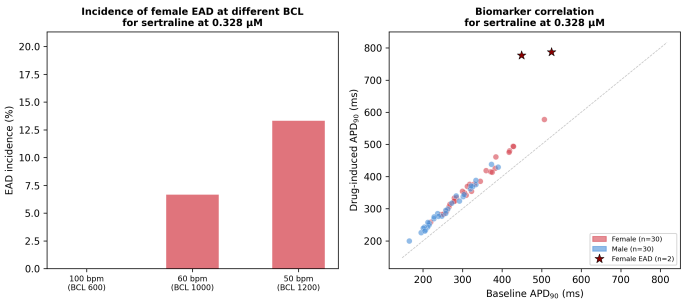

Figure 2: EAD incidence for women at varying BCL (left) and comparison of baseline and drug-induced APD_{90} (right). This reveals that I_{Kr}-blocking is riskier at rest and suggests that high baseline APD_{90} yields extra EAD risk.

Varying BCL for women, we observe that sertraline is more dangerous at rest. At slower heart rates, I_{Kr} block yields longer AP, allowing calcium channels to recover from inactivation and activate early, creating EADs [10].

The right panel of Figure 2 also lays an essential foundation: baseline and drug-induced APD_{90} correlate closely. The top 23% of female models (mapped to QTc $\geq$ 441 ms) notably show extreme prolongation ($3\times$ longer than the bottom 80%) and contain both EAD cases. This boundary can thus define a risk threshold, which, given the lack of public drug-induced ECG data, should be interpreted as an indicator rather than a validated clinical predictor.

Applying this threshold to 8,705 normal PTB-XL ECGs, 20% of women fall in the elevated-risk tier, closely matching (-3%) our simulation data, while only 14% of men fall into that same tier. Comparing both sexes across all antidepressants and dosages also confirms that women consistently experience greater QTc prolongation.

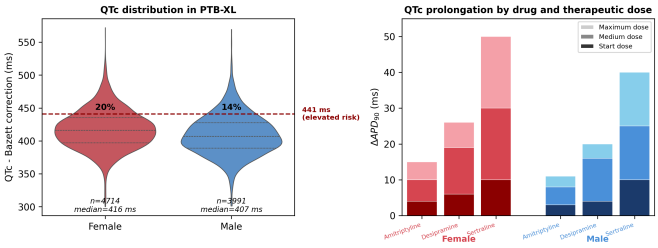

Figure 3: Violin plot distributions of QTc intervals with a simulation-derived arrhythmogenic threshold (left) and simulated QTc prolongation (right). Although QTc is an imperfect biomarker [14], it still reveals an increased risk of antidepressant-led arrhythmia for women, with sertraline appearing as the most cardiotoxic drug at any dose.

3.2. Credibility of our model and results

To verify our model, we compare our results against known behaviour: baseline APD_{90} measures match validation reports of ToR-ORd [7], and antidepressant concentrations and effects match existing literature [9, 12, 13].

We then validate our simulation against clinical data. The baseline APD_{90} sex gap matches normal ECGs (NORM confidence $\geq$ 80) in PTB-XL [11], Figure 3's QTc prolongation yields results very close to previous patient studies and align with clinical thresholds distinguishing torsadogens, and women's higher risk of drug-induced arrhythmia corroborates clinical data [8].

Finally, adding uncertainty on our most sensitive variables further supports our results. Varying IC_{50} by $\pm 25\%$ and female's I_{Kr} factor by $\pm 10\%$, ranking is preserved and women remain the only patients to trigger EAD.

3.3. Consideration of previous literature

Our findings align with sex gap reports on drug-induced QTc prolongation with I_{Kr} block [8]. Furthermore, our sex-differentiated analysis suggests that sex-neutral drug models underestimate female and overestimates male risk. Lastly, our results support the claim that antidepressants are cardiotoxic below maximum doses [3].

However, our results do not align with previous findings that TCAs carry higher arrhythmic risk than SSRIs [3]. This discrepancy likely arises because our compound-specific approach cannot generalise over three compounds.

3.4. Inconclusive and negative results

A first counter-intuitive result is that desipramine triggers no EAD at maximum dose, despite greater QTc prolongation than amitriptyline. This confirms that female vulnerability is mechanism-specific: it arises when I_{Kr} block is not counterbalanced by an inward block [14]. In ToR-ORd, I_{CaL} block reduces the depolarising plateau current needed for re-activation, allowing normal repolarisation. Female conductance scaling thus isn't enough:

drugs must leave enough inward current for EAD.

A second inconclusive result emerges from an age analysis. Since QTc sex gap is often associated with hormones [15], it should narrow after menopause, reducing drug-induced arrhythmia risks. In PTB-XL data, this age gap is insignificant, which suggests that many factors like age-related co-morbidities all contribute to QTc intervals.

4. Conclusions

4.1. Findings and contributions

This study quantifies how electrophysiological cardiac sex differences affect vulnerability to antidepressant-induced arrhythmias. Women trigger EAD at therapeutic doses whereas men do not, with sertraline posing the greatest risk, followed by amitriptyline, then desipramine. This result is driven by I_{Kr} block relative to compensatory I_{CaL}, confirming that APD_{90} prolongation alone is an insufficient arrhythmia predictor. Varying BCL, experiments further show that risk increases at rest. Mapping simulation-derived thresholds onto 8,705 normal PTB-XL ECGs, 20% of women fall into the elevated-risk category versus 14% for men. Together, these results bridge a literature gap, suggesting that sex should be considered when prescribing antidepressants with significant I_{Kr} block.

4.2. Strengths and limitations of our study

A strength of our approach is transparency: ToR-ORd isolates ion channels, enabling us to attribute sex differences directly to ionic conductance scaling rather than tissue or body level factors which are hard to isolate and interpret. The population of models further captures minority cases that average-patient simulations miss. On top of that, our pipeline is fully reproducible using documented IC_{50} values, scaling factors, and fixed random seeds.

Nonetheless, omitting tissue-level phenomena such as conduction velocity, transmural dispersion, and re-entrant circuits is also a weakness when considering clinical arrhythmia. Furthermore, mapping from APD_{90} to QTc assumes preservation from cell to heart repolarisation. Finally, our risk threshold is derived from simulations without drug-exposure validation, placing it as an indicator of relative risk rather than a clinical decision boundary.

4.3. Future work

A next step for this project would be to compare our single-cell findings with tissue and organ level simulation leveraging frameworks such as MonoAlg3D, incorporating transmural heterogeneity and fibre orientation to capture re-entrant arrhythmia mechanisms that single-cell models miss. This would strengthen clinical relevance by linking ionic-level EADs to tissue-level events like TdP.

On the clinical side, given credentialed access to drug-exposure ECG databases such as MIMIC-IV, future work could validate our findings against patient outcomes and train a machine learning classifier on baseline ECGs to flag patients at elevated risk for a given antidepressant, supporting safer personalised prescription decisions.

References

[1] Tsimpida D, Tsakiridi A, Daras K, Corcoran R, Gabbay M. Unravelling the dynamics of mental health inequalities in england. SSM Popul Health 2024;26:101669.

[2] Bogowicz P, Curtis HJ, Walker AJ, Cowen P, Geddes J et al. Trends and variation in antidepressant prescribing in english primary care. BJGP Open 2021;5(4).

[3] Zemrak WR, Kenna GA. Association of antipsychotic and antidepressant drugs with QT interval prolongation. Am J Health Syst Pharm 2008;65(11):1029–1038.

[4] Holmes M, Wang ZJ, Doste R, Camps J, Martinez-Navarro H et al. Sex-specific human electromechanical multiscale in-silico models for virtual therapy evaluation. J Mol Cell Cardiol Plus 2025;100479.

[5] Llopis-Lorente J, Gomis-Tena J, Cano J, Romero L, Saiz J et al. In silico classifiers for the assessment of drug proarrhythmicity. J Chem Inf Model 2020;60(10):5172.

[6] Muszkiewicz A, Britton OJ, Gemmell P, Passini E, Sánchez C et al. Variability in cardiac electrophysiology: using experimentally-calibrated populations of models to move beyond the single virtual physiological human paradigm. Prog Biophys Mol Biol 2016;120(1-3):115–127.

[7] Tomek J, Bueno-Orovio A, Passini E, Zhou X, Minchole A et al. Development, calibration, and validation of a novel human ventricular myocyte model in health, disease and drug block. eLife 2019;8:e48890.

[8] Vicente J, Johannesen L, Mason JW, Pueyo E, Stockbridge N et al. Sex differences in drug-induced changes in ventricular repolarization. Electrocardiol 2015;48(6):1081–1087.

[9] Lee HA, Kim KS, Hyun SA, Park SG, Kim SJ. Wide spectrum of inhibitory effects of sertraline on cardiac ion channels. Korean J Physiol Pharmacol 2012;16(5):327–332.

[10] McMillan B, Gavaghan DJ, Mirams GR. Early afterdepolarisation tendency as a simulated pro-arrhythmic risk indicator. Toxicol Res 2017;6(6):912–921.

[11] Wagner P, Strodthoff N, Bousseljot RD, Kreiseler D, Lunze FI et al. PTB-XL, a large publicly available electrocardiography dataset. Sci Data 2020;7:154.

[12] Leucht C, Huhn M, Leucht S. Amitriptyline versus placebo for major depressive disorder. Cochrane Database Syst Rev 2012;(12):CD009138.

[13] Nelson JC, Jatlow P, Quinlan DM, Bowers MB. Desipramine plasma concentration and antidepressant response. Arch Gen Psychiatry 1982;39(12):1419–1422.

[14] Grandi E, Morotti S, Pueyo E, Rodriguez B. Safety pharmacology – risk assessment: QT interval prolongation and beyond. Front Physiol 2018;9:678.

[15] Sedlak T, Shufelt C, Iribarren C, Merz CNB. Sex hormones and the QT interval. J Womens Health 2012;21(9):933–941.